

MUSA NAZIF BOYA

SOFTWARE ENGINEER (C/C++ & EMBEDDED SYSTEMS)

CONTACT

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EDUCATION

ERCIYES UNIVERSITY
Sep 2015 – Dec 2020
Bachelor of Science in **Electrical & Electronics** Engineering

REFERENCES

Metin ÖZBİLEN
+90 538 385 58 08
metinmozbilan@outlook.com
• Metin is Technical Leader of Keklikoğlu plastic company's R&D department.

Kadir ALTINDAG
+90 541 890 42 63
kadir_altindag@outlook.com
• Kadir is my Tech lead from Huawei.

SUMMARY

Result-oriented Software Engineer with over 4 years of experience specializing in C/C++ development for both Embedded Systems and Cloud/Microservices architectures. Proven track record in developing high-performance applications, from RTOS-based medical devices to large-scale media streaming (OTT) solutions. Expert in Linux environments, CMake, and system-level programming. Experienced in Agile/Scrum methodologies and open-source contributions (OpenHarmony).

TECHNICAL SKILLS

- Programming Languages: C, C++ (11/14/17), Rust, ArkTS, Bash Scripting.
- Embedded Systems: RTOS, STM32, ESP32, UART, SPI, I2C, BLE, Wi-Fi, PCB Design, Logic Analyzers.
- Cloud & Backend: Docker, Microservices Architecture, Nginx, RESTful APIs, FFmpeg.
- Streaming Protocols: HLS, DASH, MSS, SRT, RTMP.
- Tools & Databases: Git, CMake, Jenkins, Jira, SQLite, MongoDB, QT, ImGui.

PROFESSIONAL EXPERIENCE

C/C++ SOFTWARE ENGINEER

Huawei Turkey R&D Center

12.2022-Present

Key Projects:

- OpenHarmony ArkTS Compiler Project** (Jan 2025 – Present)
 - Contributing to the open-source ArkTS compiler and runtime components for the OpenHarmony OS ecosystem.
 - Developing low-level system components to optimize compiler performance and enhance runtime execution stability.
 - Focusing on system-level media capabilities and library integration using C++ and Clang.
- Cloud OTT (Over-The-Top) Media Services** (13 Months)
 - Designed and implemented microservices for a large-scale media streaming platform using C++17 and Docker.
 - Engineered low-latency streaming solutions supporting SRT, RTMP, HLS, and DASH protocols via Nginx and FFmpeg integration.
 - Developed "Hperf," an internal performance testing tool that simulates client behavior (mass user creation) to analyze stream stability and latency.
 - Integrated RESTful APIs for seamless communication between microservices and external platforms.
- WebLMT (Web-Based Local Maintenance Terminal)** (11 Months)
 - Developed remote access and management software for base stations, compatible with Linux, Windows, and RTOS.
 - Utilized C and C++ to maintain and upgrade legacy codebases, ensuring cross-platform stability.
 - Collaborated in an Agile Scrum team to deliver regular patches and feature updates aimed at improving software quality.

EMBEDDED SOFTWARE ENGINEER

Keklikoglu Plastik & SIMMOD

07.2021 - 12.2022

- Led the software and hardware design of humanoid medical simulation models used for CPR, intubation, and patient monitoring training.
- Built Bare-metal and RTOS systems on STM32 and ESP32 microcontrollers.
- Designed and implemented PCB layouts and established communication via UART, SPI, I2C, and BLE.
- Developed embedded GUIs using C and the RA8876 graphics engine.
- Performed signal processing for vital signs simulation (heart rhythm, lung sounds) and debugged hardware using oscilloscopes and logic analyzers.